

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 06

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Light: selective felling to create clearings for plants needing high light while retaining dense canopy for shade species; clearings also give warm sand / soil for lizard eggs; leave dead wood; woodland floor plants (bluebells) grow before the canopy leafs (2) • Dissolved oxygen: narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae; shallow water warms quickly for egg development (2) • Nutrients: removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground; dung patches add local nutrients (2)	6
01.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Food: wildflower grasslands for seed-eating birds; providing food species (1) • Control of predation: island nest sites unreachable by foxes; trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) (1) • Control of competitors: removing rhododendron so wildflowers get light (1) • Pollination: a range of flowering plants with different flowering times so pollinators have food all season (1) • Seed dispersal species (hippos, elephants, monkeys) conserved; control of introduced pathogens (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	

Question	Marking guidance	Marks
02.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Designation of protected areas (SSSIs, MCZs) (1) • Protection of wild birds and their nests (except some pests / game birds) (1) • Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts (1) • Uprooting wild plants generally illegal (1) • Bats must not be disturbed; roof timber treatments must be non-toxic to bats (1)	4
02.2	<i>1 mark for each point.</i> • Appendix I: species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding (1) • Examples: great apes, big cats, rhinos, blue whale (1) • Appendix II: species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected (1) • Examples: Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap (1)	4
02.3	<i>1 mark for each point.</i> • Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market (1) • Removing the profit removes the incentive to collect (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	
03.1	<i>1 mark for each point.</i> • Plants close their stomata to reduce transpiration losses (1) • Closed stomata stop the absorption of carbon dioxide so photosynthesis and growth stop (1) • Less water for nutrient absorption / transport and cell turgidity; severe shortage dehydrates cells and inhibits biochemical reactions (1)	3
03.2	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Anaerobic soil favours denitrifying bacteria and inhibits nitrifying bacteria so nitrate is lost faster than it forms; higher risk of fungal disease; root respiration inhibited (up to 2) • Drainage ditches or pipes (up to 2) • Deep ploughing; avoiding compaction by machinery / livestock; adding organic matter to encourage worms (up to 2)	4
03.3	<i>1 mark for each point.</i> • Groundwater often contains dissolved salts; when the water evaporates from the soil the salts are left behind and accumulate (1) • High soil salinity causes osmotic dehydration of roots / crop death; washing out with extra water uses more water and salinises rivers (1)	2
03.4	<i>1 mark.</i> • Add crushed lime (calcium carbonate / hydroxide); low pH increases leaching, inhibits nutrient uptake and mobilises toxic aluminium and lead (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 308-310, 313; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Cycling of materials (like biogeochemical cycles) - recycling, re-use (1) • Energy from renewable sources (1) • Human activities should support ecosystems / conserve biodiversity (1) • Separation of technical and biological materials; connected systems where the waste of one process feeds another (industrial symbiosis) (1) • Diverse systems are more resistant to change (1) • Design of products for extended use and end-of-life re-use; optimum rather than maximum production; technologies that improve efficiency and enable dismantling (1)	6
04.2	<i>1 mark for each point.</i> • Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae (1) • Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs (1)	2
04.3	<i>1 mark for each point.</i> • Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them (1) • A common approach gives a level playing field, coordinated monitoring and support for less successful countries (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	
05.1	<i>1 mark for each point.</i> • Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins); wastes of one process are raw materials for another so nothing is depleted or accumulates (1) • Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials (1) • Human systems are linear: extract - process - manufacture - use - discard; minerals are dispersed after use or mixed so recovery is difficult (1) • Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up; fossil fuel use is inherently linear and unsustainable (1)	4
05.2	<i>1 mark for each point.</i> • Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life (1) • They leach from landfill or are released by incineration; persistent, bioaccumulating (1)	2
05.3	<i>1 mark for each point.</i> • Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) (1) • Easy identification of materials (coding) and separation of components so they can be recycled (1) • Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) (1) • Use of recycled and recyclable / biodegradable materials; reduced packaging and material use (Cradle to Cradle) (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
06.1	<i>Correct answer only.</i> • B (1)	1
06.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Harvesting no faster than the MSY - resource never depleted (1) • Mixed species plantations - more food choices, relationships and niches (1) • Indigenous species - support indigenous insects, birds (1) • Mixed age structure - varied light, temperature, wind so more species; continuous cover and yield (1) • Selective logging - small clearings, less disturbance, long-term timber benefit (1) • Retention of dead wood, buffer zones, corridors (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
07.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses (1) • Setting lines at night when birds are not feeding (1) • Bird-scaring streamers / decoys (1) • Curved hooks that catch tuna but not birds (1)	3
07.2	<i>1 mark for each point.</i> • Effort: limits on boat size, engine power, days at sea, number of licences; decommissioning payments to reduce fleet size where overfishing is unavoidable (1) • Methods: bans or length limits on drift nets; bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) (1) • Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes) (1)	3
07.3	<i>1 mark for each point.</i> • Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) (1) • Juveniles are released when survival chances are good, increasing recruitment to the adult population (1)	2
07.4	<i>1 mark for each point.</i> • Egg / larval survival varies greatly between years so adult numbers do not predict recruitment (1) • Several poor years would lower the MSY; egg surveys allow catches to be reduced in advance rather than after a sudden collapse (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
08.1	<i>1 mark for each point.</i> • Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps (1) • Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade) (1)	2

Question	Marking guidance	Marks
08.2	<i>1 mark for each point.</i> • Fixed mooring buoys so boats do not anchor on coral (1) • Education of divers / snorkellers not to touch coral or kick up sediment; bans on spear fishing and souvenir collection (1) • Fishing controls: no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling (1) • Protection of turtle nesting beaches; control of coastal development; protection of mangroves and seagrass; artificial reefs (reef balls, sunk ships / oil rigs) (1)	4
08.3	<i>1 mark for each point.</i> • No sunlight so no symbiotic algae; polyps rely only on captured food so have far less energy and grow very slowly (1) • Damage from trawls takes hundreds of years to repair; the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited (1)	2
08.4	<i>1 mark for each point.</i> • Deep-water trawling (also by-catch); oil and gas exploration in deeper water; ocean acidification (1) • Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	
09.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands; HEP schemes alter river levels flooding sandbank nests of turtles / birds; species with narrow tolerance (specialists) affected first; sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade (2) • Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae (2) • Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish) (2)	6
09.2	<i>1 mark for each point.</i> • Reduces light penetration so submerged plants cannot photosynthesise (1) • Blocks the gills of filter feeders such as bivalve molluscs; sediment smothers the bed (1)	2
09.3	<i>1 mark for each point.</i> • Rare species often have very specialised habitat requirements / narrow ranges of tolerance (1) • A small change moves conditions outside their range while generalists survive; small populations cannot recover (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Succession: pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate; deflected succession and plagioclimaxes • Application: deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats); habitat creation planning the stages; controlling scrub / secondary succession; understanding that new habitats change over time • Population dynamics: birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • Application: setting sustainable harvest levels (fisheries, whaling, forestry); recognising k-selected species need strict protection and recover slowly; deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate); predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success; adaptation and range of tolerance guiding abiotic management • Evaluation: uncertainties (natural fluctuations, unknown carrying capacity); examples (Yellowstone wolves, Scottish deer, chalk grassland); judgement	9
	Model answer - Level 3 (8 marks)	
	Succession is the sequence by which pioneers colonise bare ground, modify the abiotic conditions and are replaced by later seres until a climax community controlled by climate is reached; where grazing, mowing, burning or coppicing halts it, a plagioclimax forms. This tells a conservation manager whether to intervene: chalk grassland, heathland and coppiced woodland are plagioclimaxes that disappear without continued management, whereas a climax habitat is best left alone, and habitat creation must plan the stages, control scrub and manage secondary succession.	

Question	Marking guidance	Marks
	Population dynamics - birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles and r- and k-selection - inform decisions of a different kind. Maximum sustainable yield sets harvest levels in fisheries, whaling and forestry; k-selected species that breed late and slowly need strict protection and recover slowly; culling may be justified where predators are absent, as with Scottish red deer without wolves, or where introduced species threaten natives, while the reintroduction of wolves to Yellowstone shows how restoring a predator changes vegetation through the food web. Monitoring population size, distribution, survival and age structure measures success, and knowledge of adaptation and ranges of tolerance guides how abiotic conditions are managed.	
	The limits are uncertainty: natural fluctuations, unknown carrying capacities and the difficulty of predicting how communities respond. Ecological understanding cannot remove that uncertainty, but without it conservation decisions - to manage or not, to cull or protect, how much to harvest - are guesses.	
	Weaker answer - Level 2 (5 marks)	
	Succession is when a habitat changes over time from pioneer species to a climax community. Conservationists need to understand this because some habitats like heathland will turn into woodland if they are not managed by grazing or burning. Population dynamics is about how populations grow and are limited by factors like food and predators. This helps conservation because you can work out how many animals can be taken without the population declining, and when a population needs protecting. Monitoring populations shows if conservation is working.	
	What makes the difference	
	• Level 2 explains both concepts correctly and links them to decisions, but generally; Level 3 uses the technical vocabulary (seres, plagioclimax, k-selection, MSY) with named cases (Yellowstone wolves, Scottish deer, chalk grassland).	
	• Level 3 covers more applications (habitat creation, culling introduced species, adaptation and tolerance) and evaluates the uncertainties.	
	• Structure: Level 3 gives each concept its own paragraph with the decisions it informs, then a conclusion.	
10.2	<i>1 mark.</i> • Any taxonomic group of organisms classified together by similarity (eg species, genus, family) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	

Question	Marking guidance	Marks
11.1	Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Energy flow: autotrophs and heterotrophs; losses by reflection, transmission, respiration; 100 000 to 1 000 units; few food chains beyond four levels; plant food most productive; grazing animals where crops impossible; pigs on wastes • Energetics: intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • Control of genetics: asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • GM: Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape; advantages; risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity) • Sustainability links: energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	25

Question	Marking guidance	Marks
11.2	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25